Experiment 13: Minimax Algorithm

Aim

To write a Python program to implement the Minimax algorithm for game playing.

Procedure

1. Define leaf node values.
2. Divide into MAX and MIN levels.
3. Calculate minimum values.
4. Calculate maximum value.
5. Display the optimal value.

Source Code

main.py Visualize Run ▶ Output

```
import math

scores = [3, 5, 2, 9]

def minimax(depth, node, maximizing):
    if depth == 2:
        return scores[node]

    if maximizing:
        return max(minimax(depth+1, node*2, False),
                   minimax(depth+1, node*2+1, False))
    else:
        return min(minimax(depth+1, node*2, True),
                   minimax(depth+1, node*2+1, True))

print("Optimal Value:", minimax(0, 0, True))
```

```
Optimal Value: 3
```

Result

Thus, the Minimax algorithm was implemented successfully.